

Odd End Semester Examination 2024

Name of the Program: **B.Tech. Biotechnology**

Semester: **3rd**

Name of the Course: **Cell Biology**

Course Code: **TBT 302**

Time: **3hrs**

Maximum Marks: **100**

Note:

- i. Answer **all the questions** by choosing *any two of the sub questions*.
- ii. Each question carries 20 marks

Q1	(20 marks)	CO1
(a)	Explain phase contrast and fluorescent microscopy and mention their uses.	
	OR	
(b)	Differentiate between prokaryotic and eukaryotic cells based on their structure and genome organization	
	OR	CO2
(c)	Explain the functions of ribosomes and discuss the types ribosomes found in prokaryotes and eukaryotes in detail.	
Q2	(20 marks)	
(a)	What do you understand by the fluid mosaic model of the plasma membrane? Explain in detail.	
	OR	CO3
(b)	Discuss the bioenergetics of the mitochondria and chloroplasts in plant cells.	
	OR	
(c)	Define fluorescence microscopy and electron microscopy? Describe their applications in cell biology.	
Q3	(20 marks)	CO4
(a)	Mention detail different types of Cell Adhesion Molecules found in cell with examples.	
	OR	
(b)	Explain the role of the Golgi apparatus and endoplasmic reticulum in protein sorting and post-translational modifications.	
	OR	CO5
(c)	Give diagrammatic explanation how signaling through GPCR takes place inside a cell?	
Q4	(20 marks)	
(a)	Give a detail account of how cyclins and Cdks help in the process of cell cycle regulation?	
	OR	CO5
(b)	Write short notes on i) Mitosis ii) Cell junctions	
	OR	
(c)	Evaluate the importance of the cytoskeleton and cell adhesion molecules in maintaining cell shape and enabling its functions.	
Q5	(20 marks)	CO5
(a)	Define chromosomal aberrations and provide examples of how they can lead to genetic disorders.	

OR		
(b)	Critically evaluate the role of apoptosis in cell functions and draw a schematic diagram of the apoptosis process.	
OR		
(c)	Highlight the characteristics of cancer cells and explain the molecular basis of cancer with an example.	